

Exponential inequalities

The same method as the equations, with one extra question that decides the direction of the answer.

LESSON 53–54

2 × 40 MIN

ALGEBRA 10, §3.3

P2 · 2.4

LESSON CLOCK

10'	22'	22'	24'	20'	12'
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The one extra question	10 min
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Base greater than 1	22 min
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Base less than 1	22 min
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Substitution	24 min
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Practice	20 min
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Homework	12 min
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LEARNING OBJECTIVES

- Use monotonicity to convert an exponential inequality into one about the indices.
- Reverse the inequality when the base is less than 1.
- Solve inequalities that become quadratic after a substitution.
- Write the solution in interval notation.

TEXTBOOK

National	pp. 209–220
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Cambridge	Pure Mathematics 2 & 3, pp. 39–44
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Workbook	P2 Exercise 2C
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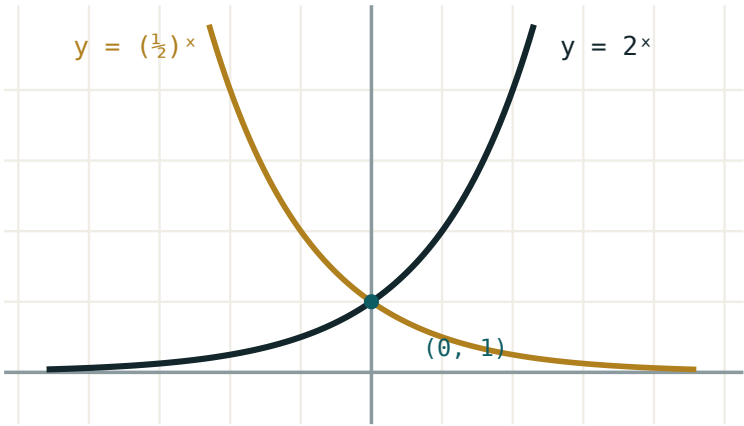
01 Explanation

The one extra question

ASK IT BEFORE ANYTHING ELSE

Is the base bigger or smaller than 1? For $a > 1$ the function increases, so the inequality between the indices points the **same** way. For $0 < a < 1$ it decreases, so the inequality **reverses**.

$$a > 1: a^m > a^n \Leftrightarrow m > n \quad 0 < a < 1: a^m > a^n \Leftrightarrow m < n$$



both pass through (0, 1); the x-axis is an asymptote

*Read it off the picture: the rising curve preserves order,
the falling curve reverses it.*

Base greater than 1

$$2^{x+1} > 8 \Rightarrow 2^{x+1} > 2^3 \Rightarrow x + 1 > 3 \Rightarrow x > 2$$

The sign is untouched. Every step is the technique of the last lesson, with > in place of =.

INEQUALITY	SAME BASE	ANSWER
$3^x > 27$	$3^x > 3^3$	$x > 3$
$2^x \leq 16$	$2^x \leq 2^4$	$x \leq 4$
$5^{2x} < 125$	$5^{2x} < 5^3$	$x < 1.5$
$4^x \geq 1$	$4^x \geq 4^0$	$x \geq 0$

Base less than 1

$$\left(\frac{1}{2}\right)^x > 8 \Rightarrow \left(\frac{1}{2}\right)^x > \left(\frac{1}{2}\right)^{-3} \Rightarrow x < -3$$

THIS IS THE MARK EVERYONE LOSES

The sign flipped, and nothing on the page says so except the base. Circle the base first; write “base < 1, so reverse” on its own line; then flip.

An alternative that avoids the trap: rewrite $\left(\frac{1}{2}\right)^x$ as 2^{-x} so the base is greater than 1, and let the minus sign do the reversing where you can see it.

Substitution

With $t = a^x > 0$ the inequality becomes an ordinary one in t , solved by the sign chart of Quarter II — but the answer must be translated back:

$$4^x - 5 \cdot 2^x + 4 < 0, t = 2^x > 0 \Rightarrow t^2 - 5t + 4 < 0 \Rightarrow 1 < t < 4$$

$$1 < 2^x < 4 \Rightarrow 0 < x < 2$$

TRANSLATE CAREFULLY

$1 < t < 4$ becomes $2^0 < 2^x < 2^2$. Because the base exceeds 1, the order survives and the answer is $0 < x < 2$. With a base under 1 the whole double inequality would reverse.

02 Worked examples

EXAMPLE 1 Solve $3^{2x-1} \geq 27$.

- ① $27 = 3^3$; base $3 > 1$.
Sign unchanged.

② $2x - 1 \geq 3$

③ $x \geq 2$

ANSWER

$$x \geq 2$$

EXAMPLE 2 Solve $(\frac{1}{3})^{(x+2)} > 9$.

① $9 = (\frac{1}{3})^{(-2)}$; base < 1 .
Reverse.

② $x + 2 < -2$

③ $x < -4$

ANSWER

$$x < -4$$

EXAMPLE 3 Solve $9^x - 4 \cdot 3^x + 3 \leq 0$.

① Let $t = 3^x > 0$.
 $t^2 - 4t + 3 \leq 0$

② $(t - 1)(t - 3) \leq 0 \Rightarrow 1 \leq t \leq 3$

③ $3^0 \leq 3^x \leq 3^1$
Base > 1 .

④ $0 \leq x \leq 1$

ANSWER

$$0 \leq x \leq 1$$

03 Interactive model

Draw the rising and falling curves side by side before solving anything.

Reading the solution set

$$x > 2$$

inequality $x > 2$ boundary **2** (open) $x = 0$ works? **no** $x = 6$ works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Exponential inequality	Ko'rsatkichli tengsizlik	Показательное неравенство
Monotonicity	Monotonlik	Монотонность
Reversing the sign	Ishorani almashtirish	Смена знака неравенства
Base greater than 1	Birdan katta asos	Основание больше единицы
Base between 0 and 1	Birdan kichik asos	Основание меньше единицы
Interval notation	Oraliq belgilanishi	Интервальная запись
Substitution	Almashtirish	Замена
Solution set	Yechimlar to'plami	Множество решений

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$2^x > 8$ gives:

$(\frac{1}{2})^x > 8$ gives:

The sign reverses when:

$1 < 2^x < 4$ gives:

$$0 < x < 2$$

$$1 < x < 4$$

$$x > 2$$

$$0 < x < 4$$

$4^x \geq 1$ gives:

$$x \geq 1$$

$$x \geq 0$$

$$x > 0$$

$$x \leq 0$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $2^x > 16$

ANSWER $x > 4$

2. Solve $3^x \leq 9$

ANSWER $x \leq 2$

3. Solve $5^x > 1$

ANSWER $x > 0$

4. Solve $(\frac{1}{2})^x > 4$

ANSWER $x < -2$

5. Solve $(\frac{1}{3})^x < 27$

ANSWER $x > -3$

6. Solve $10^x \geq 100$

ANSWER $x \geq 2$

7. Solve $2^x < 1$

ANSWER $x < 0$

Medium · the standard question

7 PROBLEMS

1. Solve $3^{2x-1} \geq 27$

ANSWER $x \geq 2$

2. Solve $(\frac{1}{3})^{x+2} > 9$

ANSWER $x < -4$

3. Solve $2^{x+1} \leq 32$

ANSWER $x \leq 4$

4. Solve $4^x > 8$

ANSWER $x > 1.5$

5. Solve $9^x \leq 3^{x+2}$

ANSWER $x \leq 2$

6. Solve $(0.2)^x \geq 25$

ANSWER $x \leq -2$

7. Solve $2^{x^2-4} < 1$

ANSWER $-2 < x < 2$

Hard · stretch and reason

7 PROBLEMS

1. Solve $9^x - 4 \cdot 3^x + 3 \leq 0$

ANSWER $0 \leq x \leq 1$

2. Solve $4^x - 5 \cdot 2^x + 4 < 0$

ANSWER $0 < x < 2$

3. Solve $25^x - 6 \cdot 5^x + 5 > 0$

ANSWER $x < 0$ or $x > 1$

4. Solve $2^{x+1} + 2^{x-1} > 20$

ANSWER $x > 3$

5. Solve $(\frac{1}{2})^{x^2} \geq \frac{1}{16}$

ANSWER $-2 \leq x \leq 2$

6. Solve $3^{x+1} < 9^{x-1}$

ANSWER $x > 3$

7. Explain why $2^x > -1$ has solution $\mathbb{R}$

ANSWER Every power of 2 is positive

07 Homework**Homework — 6 tasks**Write “base > 1 ” or “base < 1 ” on its own line before flipping anything.

1. Solve $2^{3x-2} \geq 64$.

2. Solve $(\frac{1}{4})^{x-1} > 16$.

3. Solve $9^x \leq 3^{x+6}$.

4. Solve $4^x - 10 \cdot 2^x + 16 \leq 0$.

5. Solve $3^{x^2-9} < 1$.

6. Explain in three sentences why the base decides the direction of the answer.